KCE PATHWAYS · GRADES 4–5 · UNIT 4: SHAPES AND SPACE · WEEK 20 · CHECKPOINT

Checkpoint 4 — Shapes and Space Review — Teacher Lesson Plan & Answer Key

Learning objectives: Students show mastery of Unit 4 (Weeks 16–19): dividing with unit fractions, dividing decimals, lines, rays and angles, and the volume of rectangular prisms, plus Greek and Latin roots and context clues, reading drama, opinion and argument, and theme and summary. Students also solve two-step problems, judge explanations, find mistakes, edit sentences, and write a short opinion paragraph that combines the unit's writing skills.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: the four Unit 4 math settings (Silas's apple orchard in the Blue Ridge Mountains, Leilani's farmers market in Honolulu, Imani's street survey in Washington, D.C., Makena's peach packing house in Fort Valley, Georgia) and the reading texts from Norman, Plant City, Detroit and Ocala. The student types first and last name so results are sent. Read the scoring rules together: 2 tries per question, points only for a correct first try, and the answer plus the reason appear after the 2nd miss.
2. Slide 1 — Pencil Break (Pre): the student opens or prints the Checkpoint Pre/Post sheet (grade4-5/prepost/KCE_Checkpoint_PrePost_G4-5W20.pdf) and completes Page 1 (Checkpoint 4 Warm-Up) off screen, with a protractor if available.
3. Slide 2 — Week 16 Math: Dividing with Unit Fractions, Part 1 (5 pts: 3 + 2). Each skill slide has a one-line reminder at the top; if a student misses an item, point back to it.
4. Slide 3 — Week 16 Math: Dividing with Unit Fractions, Part 2 (2 pts): a two-step divide-then-convert item.
5. Slide 4 — Week 16 Reading: Roots, Affixes, and Context Clues, Part 1 (5 pts: 3 + 2), with a short passage from Wyatt's weather log in each set.
6. Slide 5 — Week 16 Reading: Spell It and Punctuate It, Part 2 (2 pts): fix a suffix spelling AND the capitals and punctuation; checked exactly (with or without the comma after an introductory phrase where shown).
7. Slide 6 — Week 17 Math: Dividing Decimals, Part 1 (5 pts: 3 + 2).
8. Slide 7 — Week 17 Math: Dividing Decimals, Part 2 (2 pts): estimate first, then choose.
9. Slide 8 — Week 17 Reading: Reading Drama, Part 1 (5 pts: 3 + 2), with a new Scene 3 from Frost Watch in each set.
10. Slide 9 — Week 17 Reading: Script to Story, Part 2 (2 pts): turn a script line into a story sentence; checked exactly.
11. Slide 10 — Week 18 Math: Lines, Rays, and Angles, Part 1 (5 pts: 3 + 2).
12. Slide 11 — Week 18 Math: Angles as Turns, Part 2 (2 pts): a fraction of a 360° turn or a clock angle.
13. Slide 12 — Week 18 Reading: Opinion and Argument, Part 1 (5 pts: 3 + 2), with a short letter about a Detroit garden in each set.
14. Slide 13 — Week 18 Reading: Commas Every Arguer Needs, Part 2 (2 pts): introductory-word and series commas; checked exactly (series with or without the comma before and).

15. Slide 14 — Week 19 Math: Volume of Rectangular Prisms, Part 1 (5 pts: 3 + 2).
16. Slide 15 — Week 19 Math: Packing Problems, Part 2 (2 pts): a composite figure or packing item.
17. Slide 16 — Week 19 Reading: Theme and Summary, Part 1 (5 pts: 3 + 2), with a fable, story excerpt or short poem in each set.
18. Slide 17 — Week 19 Reading: Titles and Dialogue, Part 2 (2 pts): titles in quotation marks with commas and periods inside; checked exactly.
19. Slide 18 — Putting Skills Together, Part 1 (6 pts, 3 each): a two-step problem that mixes two weeks and a "which explanation is correct?" item.
20. Slide 19 — Be the Checker, Part 2 (6 pts, 3 each): a math "find the mistake" and a reading/writing "find the mistake." Ask the student to say why each wrong choice is wrong.
21. Slide 20 — Fix the Sentence, Part 1 (4 pts, 2 each): typed sentences checked exactly (capital letters, quotation marks, commas, end marks, its/it's). Straight or curly quotes and extra spaces are OK.
22. Slide 21 — Fix the Sentence, Part 2 (4 pts, 2 each): find the subject before choosing the verb; one item accepts either verb tense.
23. Slide 22 — Writer's Notebook: Make Your Case: an opinion paragraph of 3 to 5 sentences about the Unit 4 place the student would most like to visit (practice, XP only), then one scored follow-up item (4 pts). Read the notebook writing aloud together if time allows.
24. Slide 23 — Match-Up Round (10 pts): 5 Unit 4 vocabulary words across math and reading, 2 pts each on the first try.
25. Slide 24 — Looking Back on Unit 4 reflection (bonus XP only).
26. Slide 25 — Unit 4 Boss Challenge (10 pts, 5 each): a multi-step math problem with a different situation in each set and a reading/writing item that combines two skills.
27. Slide 26 — Pencil Break (Post): the student completes Page 2 (Checkpoint 4 Challenge) of the same sheet (grade4-5/prepost/KCE_Checkpoint_PrePost_G4-5W20.pdf), plus the ★ Challenge if time allows, before seeing the score.
28. Slide 27 — Results (Unit 4 Complete!): the score out of 100 is sent to KingCretot Experience, with a score breakdown by week so you can see what to review. 85 or higher shows "Unit 4: Shapes and Space Complete!" with the congratulations message "You did it! Unit 4 is complete!" and a big confetti burst; below 85 shows "Good Work on Checkpoint 4!" with encouragement ("You are getting closer every time."), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Checkpoint 4 Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

DIVIDING WITH UNIT FRACTIONS (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
a1_s0	Set 1: *Silas's Orchard, Revisited Silas is back at his family's apple orchard near Hendersonville, North Carolina, helping Ms. Dunbar in the orchard kitchen and Mr. Barrera at the packing table.* Ms. Dunbar has 1/4 of a pan of apple bread left. She cuts it into 6 equal pieces. What fraction of the WHOLE pan is each piece? 1/4 ÷ 6 =	**1/24** (also accepted: 1/24 of the pan, 1/24 pan, 1/24 of a pan)	Splitting 1/4 into 6 equal pieces makes twenty-fourths of the whole pan, because 4 × 6 = 24. So 1/4 ÷ 6 = 1/24. Check: 1/24 × 6 = 6/24 = 1/4.

ID	Question	Answer	Why (explain to student)
a2_s0	Set 1: Mr. Barrera has 6 pounds of dried apple chips. He packs them into 1/3-pound bags. How many bags can he fill? Bags:	**18** (also accepted: 18 bags, 18 bag)	Each pound holds 3 bags of 1/3 pound, so 6 pounds hold 6 × 3 = 18 bags. 6 ÷ 1/3 = 18. Check: 18 × 1/3 = 18/3 = 6.
a3_s0	Set 1: Two steps: Silas splits 1/4 pound of cinnamon equally into 2 shakers. How many OUNCES go in each shaker? (1 pound = 16 ounces) Ounces:	**2** (also accepted: 2 ounces, 2 ounce, 2 oz)	Step 1: 1/4 ÷ 2 = 1/8 pound in each shaker. Step 2: 1/8 of 16 ounces is 16 ÷ 8 = 2 ounces.
a1_s1	Set 2: *Silas's Orchard, Revisited Silas is back at the orchard near Hendersonville, North Carolina, filling baskets and tying bows for Grandma Lucinda's farm stand.* Silas has 1/3 of a crate of apples. He splits it equally into 4 baskets. What fraction of the WHOLE crate goes in each basket? 1/3 ÷ 4 =	**1/12** (also accepted: 1/12 of the crate, 1/12 crate, 1/12 of a crate)	Splitting 1/3 into 4 equal parts makes twelfths of the whole crate, because 3 × 4 = 12. So 1/3 ÷ 4 = 1/12. Check: 1/12 × 4 = 4/12 = 1/3.
a2_s1	Set 2: Grandma Lucinda has 4 yards of ribbon. Each bow uses 1/5 yard. How many bows can she tie? Bows:	**20** (also accepted: 20 bows, 20 bow)	Each yard holds 5 pieces of 1/5 yard, so 4 yards hold 4 × 5 = 20 bows. 4 ÷ 1/5 = 20. Check: 20 × 1/5 = 20/5 = 4.
a3_s1	Set 2: Two steps: Ms. Dunbar has 1/3 hour to frost 4 cakes, with the same amount of time for each cake. How many MINUTES does each cake get? Minutes:	**5** (also accepted: 5 minutes, 5 minute, 5 min)	Step 1: 1/3 ÷ 4 = 1/12 hour for each cake. Step 2: 1/12 of 60 minutes is 60 ÷ 12 = 5 minutes.
a1_s2	Set 3: *Silas's Orchard, Revisited Silas is back at the orchard near Hendersonville, North Carolina, getting jars, pies, and chip bags ready for the North Carolina Apple Festival.* Grandma Lucinda pours 1/5 gallon of apple butter equally into 3 jars. How much of a gallon goes in each jar? Gallon in each jar:	**1/15** (also accepted: 1/15 gallon, 1/15 gal, 1/15 of a gallon, 1/15 gallons)	Splitting 1/5 into 3 equal parts makes fifteenths, because 5 × 3 = 15. So 1/5 ÷ 3 = 1/15 gallon. Check: 1/15 × 3 = 3/15 = 1/5.
a2_s2	Set 3: Ms. Dunbar bakes 3 apple pies and cuts every pie into sixths. How many slices of 1/6 pie does she have? Slices:	**18** (also accepted: 18 slices, 18 slice, 18 pieces)	Each pie holds 6 sixths, so 3 pies hold 3 × 6 = 18 slices. 3 ÷ 1/6 = 18. Check: 18 × 1/6 = 18/6 = 3.
a3_s2	Set 3: Two steps: Mr. Barrera packs 2 pounds of apple chips into 1/4-pound bags. He sells 3 bags before lunch. How many bags are LEFT? Bags left:	**5** (also accepted: 5 bags, 5 bag)	Step 1: 2 ÷ 1/4 = 2 × 4 = 8 bags. Step 2: he sells 3, so 8 − 3 = 5 bags are left.

ROOTS, AFFIXES, AND CONTEXT CLUES (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: *From Wyatt's weather log, Norman, Oklahoma 1The morning after the storm, Noor read the hygrometer on the shed wall. The air was still humid, or full of water vapor. 2Yesterday Mr. Harjo's voice on the radio was inaudible. Today every word was audible, loud and clear.* In paragraph 1, which kind of context clue helps you figure out humid? [A synonym or restatement clue, after the word or / An antonym clue, near the word but / An example clue, with a list of examples / There is no clue in the paragraph]	**A synonym or restatement clue, after the word or**	The word or comes right after humid and restates its meaning: "full of water vapor." That is a synonym or restatement clue.
b2_s0	Set 1: Type your answer: audible and inaudible share the Latin root aud. The root aud means ___. aud means:	**hear** (also accepted: to hear, hearing)	The Latin root aud means hear, as in audience and inaudible. Audible means able to be heard.
b1_s1	Set 2: *From Wyatt's weather log, Norman, Oklahoma 1Ms. Nwosu packed a portable radio in her storm bag. The big station radio stays bolted to the wall, but this one can go anywhere. 2By evening, a thin, almost invisible mist hung over the pond.* Two steps: what does portable mean, AND which clue in paragraph 1 helps? [It can be carried; the contrast with the radio bolted to the wall helps / It is very loud; the word storm helps / It cannot be carried; the word but means the same thing / It is broken; the word bag gives an example]	**It can be carried; the contrast with the radio bolted to the wall helps**	port means carry and -able means can be, so portable means it can be carried. The contrast with the big radio that stays bolted to the wall (signaled by but) confirms it.
b2_s1	Set 2: Type your answer: invisible = in- (not) + ___ (see) + -ible (can be). Type the Latin root that means see. Root:	**vis** (also accepted: vis-, v-i-s)	invisible = in- (not) + vis (see) + -ible (can be). The root that means see is vis.
b1_s2	Set 3: *From Wyatt's weather log, Norman, Oklahoma 1By evening, the wind howled so loudly that it seemed to interrupt every sentence Wyatt tried to say. His words broke off in the middle. 2Ms. Nwosu tapped the barometer and smiled. "Rising pressure," she said. "Tomorrow will be fair."* Interrupt = inter- (between) + rupt (break). Which meaning fits paragraph 1? [To break in between, so a sentence stops in the middle / To carry something across a long distance / To look closely at something / To say what will happen before it happens]	**To break in between, so a sentence stops in the middle**	inter- means between and rupt means break. The wind broke in between Wyatt's words, so they "broke off in the middle." That is an inference clue from what happens.
b2_s2	Set 3: Type your answer: add the prefix that means "not" to the word legal. Choose the right spelling of the prefix and type the whole new word. New word:	**illegal**	The prefix in- (not) becomes il- before a word that starts with l. Keep the l in legal, so the word has a double l: illegal.

SPELL IT AND PUNCTUATE IT (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b3_s0	Set 1: Fix this sentence: after the storm the creek road was unuseable	**After the storm, the creek road was unusable.** (also accepted: After the storm the creek road was unusable.)	Capitalize the first word and add a comma after the opening phrase After the storm. Use + able drops the silent e, because -able starts with a vowel: unusable.
b3_s1	Set 2: Fix this sentence: noor was hopefull that the storm would pass	**Noor was hopeful that the storm would pass.**	Capitalize the name Noor, which starts the sentence. The suffix -ful has only one l: hope + ful = hopeful. End with a period.
b3_s2	Set 3: Fix this sentence: the rain gauge was relyable all spring	**The rain gauge was reliable all spring.**	Capitalize the first word. When a consonant comes before the y, change y to i before adding -able: rely + able = reliable. End with a period.

DIVIDING DECIMALS (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
c1_s0	Set 1: *Leilani's Market Morning, Revisited Leilani is back at the Saturday farmers market in Kaka'ako, Honolulu, helping her grandmother, Tutu Mahina, and their neighbor Mr. Haddad at the juice tent.* Mr. Haddad pours 8.4 liters of guava juice equally into 6 pitchers. How many liters go in each pitcher? Liters:	**1.4** (also accepted: 1.40, 1.4 liters, 1.4 liter, 1.4 L)	8 ones ÷ 6 = 1 one with 2 ones left. 2 ones = 20 tenths, plus 4 tenths = 24 tenths, and 24 tenths ÷ 6 = 4 tenths. So 8.4 ÷ 6 = 1.4. Check: 1.4 × 6 = 8.4.
c2_s0	Set 1: Type the quotient. Fill any empty places with zeros. 5.2 ÷ 100 =	**0.052** (also accepted: .052)	Dividing by 100 moves every digit two places to the right: the 5 moves from ones to hundredths and the 2 from tenths to thousandths. A 0 holds the tenths place. 5.2 ÷ 100 = 0.052.
c3_s0	Set 1: A box of 4 papayas costs $9.80. Estimate first, then choose the unit price of ONE papaya. [$2.45 / $24.50 / $0.245 / $2.20]	**$2.45**	$9.80 ÷ 4: 9 ÷ 4 = 2 with 1 left; 18 tenths ÷ 4 = 4 tenths with 2 tenths left; 20 hundredths ÷ 4 = 5 hundredths. So $2.45. Check: $2.45 × 4 = $9.80. Estimate: $8 ÷ 4 = $2, so the answer is a little more than $2.
c1_s1	Set 2: *Leilani's Market Morning, Revisited Leilani is back at the farmers market in Honolulu, cutting ribbon for gift boxes and helping her cousin Caleb label the fruit.* Write a zero to keep dividing. Type the exact quotient. 7.8 ÷ 4 =	**1.95**	7.8 ÷ 4 = 1.9 with 2 tenths left over. Write 7.8 as 7.80 and keep dividing: 20 hundredths ÷ 4 = 5 hundredths. So 7.8 ÷ 4 = 1.95. Check: 1.95 × 4 = 7.80.
c2_s1	Set 2: A roll of ribbon for the fruit stand is 275 centimeters long. How many METERS long is it? Meters:	**2.75** (also accepted: 2.75 meters, 2.75 meter, 2.75 m)	1 meter = 100 centimeters. Centimeters are smaller, so divide by 100: 275 ÷ 100 = 2.75 meters.

ID	Question	Answer	Why (explain to student)
c3_s1	Set 2: Watch for the placeholder zero. Which is the correct quotient for 0.54 ÷ 6? [0.09 / 0.9 / 9 / 0.009]	**0.09**	0.54 is 54 hundredths. 54 hundredths ÷ 6 = 9 hundredths = 0.09. The 0 in the tenths place is a placeholder. Check: 0.09 × 6 = 0.54.
c1_s2	Set 3: *Leilani's Market Morning, Revisited Leilani is back at Tutu Mahina's fruit stand in Honolulu, pricing mangoes and weighing macadamia nuts.* A bag of 6 mangoes costs $13.50. What is the unit price, the cost of ONE mango? Dollars: $	**2.25** (also accepted: $2.25, 2.25 dollars)	13 ÷ 6 = 2 with 1 left over. 1 one and 5 tenths make 15 tenths, and 15 tenths ÷ 6 = 2 tenths with 3 tenths left. 30 hundredths ÷ 6 = 5 hundredths. $13.50 ÷ 6 = $2.25. Check: $2.25 × 6 = $13.50.
c2_s2	Set 3: A sack of macadamia nuts has a mass of 1,240 grams. What is its mass in KILOGRAMS? Kilograms:	**1.24** (also accepted: 1.24 kilograms, 1.24 kilogram, 1.24 kg)	1 kilogram = 1,000 grams. Grams are smaller, so divide by 1,000: 1,240 ÷ 1,000 = 1.24 kilograms. That is a little more than 1 kilogram, which makes sense.
c3_s2	Set 3: Find the missing divisor: 48.6 ÷ ___ = 0.486 [100 / 10 / 1,000 / 6]	**100**	Every digit moved two places to the right (the 4 went from tens to tenths), so the divisor is 100. 48.6 ÷ 100 = 0.486.

READING DRAMA (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
d1_s0	Set 1: *From Frost Watch, a play by Lorena Vance • a new scene (Scene 3. The strawberry field at sunrise. Ice glitters on every plant. ABUELA CARMEN walks slowly down a row.) TEO: (yawning) Is it over? Did we win? ABUELA CARMEN: (kneeling, touching a flower) Look. The flowers are still white inside. PALOMA: (writing fast in her notebook) Thirty-one degrees at six o'clock. I'm writing it all down for next year.* What do Paloma's line and the stage direction (writing fast in her notebook) reveal about her? [She keeps careful records so she can plan ahead / She is bored and wants to go back to bed / She does not care what happens to the farm / She is making a joke to cheer up Teo]	**She keeps careful records so she can plan ahead**	Paloma writes the temperature and time in her notebook "for next year." Her words and her action (a stage direction) show she keeps careful records to plan ahead.
d2_s0	Set 1: Type the missing word: the list at the very beginning of a play that names every character is the cast of ___. Cast of:	**characters** (also accepted: character, cast of characters)	A play begins with the cast of characters: the list that names everyone in the play, often with a short description of each one.

ID	Question	Answer	Why (explain to student)
d1_s1	Set 2: *From Frost Watch, a play by Lorena Vance • a new scene (Scene 3. The strawberry field at sunrise. MR. PRYOR walks over from his truck.) MR. PRYOR: (gruffly, handing PALOMA a thermos) Hot cocoa. Don't tell anybody I'm nice. PALOMA: (smiling) Your secret is safe with me, Mr. Pryor.* What do Mr. Pryor's line and stage direction reveal about him? [He acts gruff, but his actions show he is really kind / He is angry that the pump broke / He wants Paloma to pay for the cocoa / He does not like the Delgado family]	**He acts gruff, but his actions show he is really kind**	He speaks gruffly, but his action (bringing hot cocoa) is kind, and his joke shows he does not want anyone to know. Paloma's smile shows she sees his kindness.
d2_s1	Set 2: Type the missing word: (gruffly, handing PALOMA a thermos) is not spoken out loud. Words like these are called stage ___. Stage:	**directions** (also accepted: direction, stage directions)	Words in parentheses and italics that tell actors how to move or speak are stage directions. Actors do not say them out loud.
d1_s2	Set 3: *From Frost Watch, a play by Lorena Vance • a new scene (Scene 3. The Delgado kitchen, late morning. The sprinklers are finally off.) ABUELA CARMEN: (wiping her eyes) Forty years on this land, and the neighbors still come running. TEO: (puffing out his chest) Because I ran the fastest!* A narrator can tell feelings directly. Which prose sentence best tells what (wiping her eyes) shows about Abuela Carmen? [Abuela Carmen felt so thankful that tears came to her eyes. / ABUELA CARMEN: (wiping her eyes) I am thankful. / Abuela Carmen was angry at the neighbors. / Abuela Carmen had something in her eye from the dust.]	**Abuela Carmen felt so thankful that tears came to her eyes.**	A narrator can TELL a feeling that a stage direction only SHOWS. Abuela wipes her eyes while speaking about neighbors who helped, so she is moved and thankful.
d2_s2	Set 3: Type the word, and spell it carefully: a person who writes plays, like Lorena Vance, is a ___. Word:	**playwright** (also accepted: a playwright, playwrights)	A person who writes plays is a playwright. It ends in -wright, an old word for maker.

SCRIPT TO STORY (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
d3_s0	Set 1: Script to story: the script says TEO: Is it over? Fix the story sentence below with capital letters, a comma, quotation marks, and the end mark. teo asked is it over	**Teo asked, "Is it over?"**	In a story, Teo's exact words go inside quotation marks with a comma after the tag. Capitalize Teo and Is. His words are a question, so the question mark goes inside the closing quotation mark.
d3_s1	Set 2: Script to story: the script says PALOMA: Your secret is safe with me, Mr. Pryor. Fix the story sentence below. Paloma said your secret is safe with me Mr. Pryor	**Paloma said, "Your secret is safe with me, Mr. Pryor."**	Put a comma after the tag, quotation marks around Paloma's exact words, and a capital Y. Paloma speaks TO Mr. Pryor, so a comma sets off his name (direct address). The period goes inside the closing quotation mark.

ID	Question	Answer	Why (explain to student)
d3_s2	Set 3: Script to story: the script says TEO: Because I ran the fastest! Fix the story sentence below. teo shouted because i ran the fastest	**Teo shouted, "Because I ran the fastest!"**	Capitalize Teo, Because, and I. Put a comma after the tag and quotation marks around Teo's exact words. His words are an exclamation, so the exclamation point goes inside the closing quotation mark.

LINES, RAYS, AND ANGLES (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: *Imani's Street Survey, Revisited Imani is back in Washington, D.C., measuring street corners with her uncle, Mr. Adeyemi, the land surveyor, and Ms. Ruiz, the street designer.* A diagonal avenue meets a straight street. The two angles on one side of the street sit on a straight line. One is 117°. What is the other angle? Degrees:	**63** (also accepted: 63°, 63 °, 63 degrees, 63 degree, 63 deg)	The two angles make a straight angle, which is 180°. 180 − 117 = 63°. Check: 117 + 63 = 180.
e2_s0	Set 1: Which list shows an ACUTE angle, a RIGHT angle, and an OBTUSE angle, in that order? [40°, 90°, 160° / 90°, 40°, 160° / 40°, 160°, 90° / 40°, 90°, 180°]	**40°, 90°, 160°**	Acute is less than 90°, right is exactly 90°, and obtuse is more than 90° but less than 180°. 40°, 90°, 160° fits that order. 180° is a straight angle, not obtuse.
e1_s1	Set 2: *Imani's Street Survey, Revisited Imani is back in Washington, D.C., marking lines and corners on Mr. Adeyemi's street map.* The square in a corner shows a right angle. A ray splits it into two parts. One part is 18°. What is the other part? Degrees:	**72** (also accepted: 72°, 72 °, 72 degrees, 72 degree, 72 deg)	A right angle is 90°. 90 − 18 = 72°. Check: 18 + 72 = 90.
e2_s1	Set 2: On the map, line m and line n never meet. Line p crosses both of them at square corners. Which statement is true? [m and n are parallel, and p is perpendicular to both / m and n are perpendicular, and p is parallel to both / m, n, and p are all parallel / p is a ray, so it cannot cross m or n]	**m and n are parallel, and p is perpendicular to both**	Lines that never meet are parallel. A line that crosses them at square corners (right angles) is perpendicular to them.
e1_s2	Set 3: *Imani's Street Survey, Revisited Imani is back in Washington, D.C., helping Ms. Ruiz plan paths around a fountain and label the map.* Three paths meet at a fountain and make three angles all the way around it. Two of the angles are 150° and 95°. What is the third angle? Degrees:	**115** (also accepted: 115°, 115 °, 115 degrees, 115 degree, 115 deg)	All the way around a point is 360°. 150 + 95 = 245, and 360 − 245 = 115°. Check: 150 + 95 + 115 = 360.
e2_s2	Set 3: A figure starts at point R and goes on forever through point S. What is its name? [ray RS / ray SR / line segment RS / point RS]	**ray RS**	A ray has one endpoint and goes on forever in one direction. Name it endpoint first: ray RS.

ANGLES AS TURNS (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e3_s0	Set 1: Two steps: Mr. Adeyemi's watch shows exactly 5:00. What is the smaller angle between the hour hand and the minute hand? Degrees:	**150** (also accepted: 150°, 150 °, 150 degrees, 150 degree, 150 deg)	Step 1: a clock face is 360° split into 12 spaces, so each space is 360 ÷ 12 = 30°. Step 2: at 5:00 the hands are 5 spaces apart: 5 × 30 = 150°.
e3_s1	Set 2: Two steps: a cyclist turns 1/2 of a full turn at one corner, then 1/8 of a full turn at the next corner. How many degrees did she turn in all? Degrees:	**225** (also accepted: 225°, 225 °, 225 degrees, 225 degree, 225 deg)	Step 1: 1/2 turn = 360 ÷ 2 = 180°, and 1/8 turn = 360 ÷ 8 = 45°. Step 2: 180 + 45 = 225°.
e3_s2	Set 3: Two steps: a robot street sweeper makes 3 equal turns that add up to exactly a 1/4 turn. How many degrees is each of the 3 turns? Degrees:	**30** (also accepted: 30°, 30 °, 30 degrees, 30 degree, 30 deg)	Step 1: a 1/4 turn is 360 ÷ 4 = 90°. Step 2: three equal turns make 90°, so each one is 90 ÷ 3 = 30°.

OPINION AND ARGUMENT (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
f1_s0	Set 1: *From a letter to the city council by Rosa Ortega, Linden Street neighbor, Detroit 1The lot on Linden Street should become a garden with a small play area. First, children on our block have no safe place to play. The nearest park is across a busy six-lane road. 2Some people say a garden will cost too much. However, a hardware store has offered to donate soil, seeds, and tools.* Which sentence is EVIDENCE for the reason in paragraph 1? [The nearest park is across a busy six-lane road. / The lot on Linden Street should become a garden with a small play area. / First, children on our block have no safe place to play. / Some people say a garden will cost too much.]	**The nearest park is across a busy six-lane road.**	The fact about the park across a six-lane road can be checked, and it proves the reason that children have no safe place to play. The other choices are the claim, the reason, and the counterclaim.
f2_s0	Set 1: Type one word: in paragraph 2, the sentence that begins with However answers the other side. It is called the ___. Word:	**rebuttal** (also accepted: a rebuttal, the rebuttal, rebuttals)	Paragraph 2 states the other side (cost), then the However sentence answers it with a checkable fact. An answer to a counterclaim is a rebuttal.

ID	Question	Answer	Why (explain to student)
f1_s1	Set 2: *From a letter to the city council by Dev Mehta, owner of a bakery on Linden Street, Detroit 1The city should split the lot: half garden and half parking. Last month, 52 customers told me they could not find a place to park near my bakery. 2Some neighbors say any parking will ruin the garden. However, a split lot would still leave room for 20 garden beds. Bakeries are the heart of every street.* Which sentence is an OPINION that cannot be checked? [Bakeries are the heart of every street. / Last month, 52 customers told me they could not find a place to park near my bakery. / However, a split lot would still leave room for 20 garden beds. / The city should split the lot: half garden and half parking.]	**Bakeries are the heart of every street.**	"Bakeries are the heart of every street" tells what the writer feels. It cannot be checked, so it is an opinion, not evidence. The customer count and the 20 garden beds can be checked.
f2_s1	Set 2: Type one word: "Some neighbors say any parking will ruin the garden" tells the other side's opinion. It is the ___. Word:	**counterclaim** (also accepted: counter claim, counter-claim, counterargument, counter argument, counter-argument, opposing claim)	This sentence tells the other side's opinion, so it is the counterclaim. The writer states it, then answers it.
f1_s2	Set 3: *From a letter to the principal by Mrs. Okafor's fifth-grade class, Ash Avenue Elementary, Detroit 1Our school should build a garden behind the gym. Last year, the 24 students in our garden club grew 150 pounds of tomatoes and beans in borrowed pots. 2Some parents worry the garden will take time away from math. However, students measure, weigh, and count every harvest.* Which is the STRONGEST evidence for the class's claim? [Last year, the 24 students in our garden club grew 150 pounds of tomatoes and beans. / Gardens are the best part of any school. / We really, really want a garden. / Our school has a gym.]	**Last year, the 24 students in our garden club grew 150 pounds of tomatoes and beans.**	The garden club's 24 students and 150 pounds can be checked, and they show students can really grow food. The other choices are feelings, or a true fact that does not prove the claim.
f2_s2	Set 3: Type one word: a letter that tries to convince the principal to agree and to act is trying to ___ her. Word:	**persuade** (also accepted: to persuade, convince)	Both letters try to persuade: to convince the reader to agree or to act.

COMMAS EVERY ARGUER NEEDS (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
f3_s0	Set 1: Fix this sentence: Second the garden would need water hoses rakes and buckets.	**Second, the garden would need water hoses, rakes, and buckets.** (also accepted: Second, the garden would need water hoses, rakes and buckets.)	Put a comma after the introductory word Second. Water hoses, rakes, and buckets are three items in a series, so a comma goes after each item except the last.
f3_s1	Set 2: Fix this sentence: In addition the lot is close to the bus stop.	**In addition, the lot is close to the bus stop.**	In addition is an introductory phrase, so a comma follows it.

ID	Question	Answer	Why (explain to student)
f3_s2	Set 3: Fix this sentence: However students measure weigh and count every harvest.	**However, students measure, weigh, and count every harvest.** (also accepted: However, students measure, weigh and count every harvest.)	However is an introductory word, so a comma follows it. Measure, weigh, and count are three items in a series, so a comma goes after each item except the last.

VOLUME OF RECTANGULAR PRISMS (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
g1_s0	Set 1: *Makena's Packing House, Revisited Makena is back at the peach packing house in Fort Valley, Georgia, where her Uncle Thaddeus is the packing manager.* A peach gift box is 9 inches long, 4 inches wide, and 5 inches tall. What is its volume in cubic inches? Cubic inches:	**180** (also accepted: 180 cubic inches, 180 cubic inch, 180 cubic in, 180 cu in, 180 in3, 180 in³, 180 cubic units)	V = l × w × h = 9 × 4 × 5. 9 × 4 = 36, and 36 × 5 = 180 cubic inches.
g2_s0	Set 1: Two steps: a cooler holds 288 cubic feet. It is 8 feet long and 6 feet wide. How tall is it? Height (feet):	**6** (also accepted: 6 feet, 6 foot, 6 ft)	Base area: 8 × 6 = 48 square feet. Height: 288 ÷ 48 = 6 feet. Check: 8 × 6 × 6 = 288.
g1_s1	Set 2: *Makena's Packing House, Revisited Makena is back at the packing house in Fort Valley, Georgia, helping Ms. Lindqvist plan the cold storage rooms.* The base of a box of peach jam has an area of 35 square inches. The box is 8 inches tall. What is its volume in cubic inches? Cubic inches:	**280** (also accepted: 280 cubic inches, 280 cubic inch, 280 cubic in, 280 cu in, 280 in3, 280 in³, 280 cubic units)	V = B × h = 35 × 8 = 280 cubic inches.
g2_s1	Set 2: A cube-shaped storage bin has edges that are each 5 feet long. What is its volume in cubic feet? Cubic feet:	**125** (also accepted: 125 cubic feet, 125 cubic foot, 125 cubic ft, 125 cu ft, 125 ft3, 125 ft³, 125 cubic units)	Every edge of a cube is the same: 5 × 5 × 5. 5 × 5 = 25, and 25 × 5 = 125 cubic feet.
g1_s2	Set 3: *Makena's Packing House, Revisited Makena is back at the packing house in Fort Valley, Georgia, stacking candy cubes and measuring sheds.* Two steps: Makena fills a box with unit cubes. The bottom layer is 6 cubes long and 5 cubes wide, and the box holds 4 layers. What is its volume in cubic units? Cubic units:	**120** (also accepted: 120 cubic units, 120 cubic unit, 120 cubes, 120 cube, 120 unit cubes, 120 units3, 120 units³)	Bottom layer: 6 × 5 = 30 cubes. Four layers: 30 × 4 = 120 cubes, so 120 cubic units.
g2_s2	Set 3: Two steps: a composite shed has a total volume of 300 cubic feet. Prism A is 6 feet long, 5 feet wide, and 4 feet tall. What is the volume of Prism B? Cubic feet:	**180** (also accepted: 180 cubic feet, 180 cubic foot, 180 cubic ft, 180 cu ft, 180 ft3, 180 ft³, 180 cubic units)	Prism A: 6 × 5 × 4 = 120 cubic feet. Prism B: 300 − 120 = 180 cubic feet.

PACKING PROBLEMS (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
g3_s0	Set 1: Loading dock steps are made of two prisms. Prism A is 7 feet by 3 feet by 2 feet. Prism B is 4 feet by 3 feet by 1 foot. What is the total volume? [54 cubic feet / 504 cubic feet / 20 cubic feet / 42 cubic feet]	**54 cubic feet**	Prism A: 7 × 3 × 2 = 42. Prism B: 4 × 3 × 1 = 12. Add the volumes: 42 + 12 = 54 cubic feet. (504 multiplies the volumes, and 20 adds all six edges.)
g3_s1	Set 2: Two steps: a space in the truck is 10 feet long, 4 feet wide, and 6 feet tall. Crates are cubes with 2-foot edges. How many crates fit? [30 crates / 240 crates / 60 crates / 15 crates]	**30 crates**	Fit the crates one direction at a time: 10 ÷ 2 = 5 long, 4 ÷ 2 = 2 wide, 6 ÷ 2 = 3 tall. 5 × 2 × 3 = 30 crates. Check: 240 cubic feet ÷ 8 cubic feet per crate = 30. (240 is the volume of the space, not the number of crates.)
g3_s2	Set 3: A box is 3 inches long, 5 inches wide, and 4 inches tall. Uncle Thaddeus doubles only the length. What is the new volume? [120 cubic inches / 240 cubic inches / 70 cubic inches / 64 cubic inches]	**120 cubic inches**	The box starts at 3 × 5 × 4 = 60 cubic inches. Doubling only the length gives 6 × 5 × 4 = 120 cubic inches, which is double. (70 comes from adding 60 + 10.)

THEME AND SUMMARY (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
h1_s0	Set 1: *"The Oak and the Reeds," an Aesop fable, retold* *A tall Oak grew beside a river, where thin Reeds grew in the mud. "Why do you bow to every little breeze?" the Oak asked. "I never bend." That night a great storm came. The Reeds bent low and let the wind pass over them. The proud Oak stood stiff against the wind until it cracked and crashed into the river. Moral: It is better to bend than to break.* Which is a THEME (a full-sentence message about life), not a topic, an event, or a command? [Being flexible can help you get through hard times. / Trees / The oak fell into the river. / Be like a reed.]	**Being flexible can help you get through hard times.**	A theme is a full-sentence message about life that is true outside the story. "Trees" is only a topic, the sentence about the oak falling is an event, and "Be like a reed" is a command.
h2_s0	Set 1: Type one word: a fable, a poem, and a realistic story are three different ___, or kinds of writing. Word:	**genres** (also accepted: genre, kinds of writing)	A genre is a kind of writing. Fables, poems, and realistic stories are three different genres.

ID	Question	Answer	Why (explain to student)
h1_s1	Set 2: *From a new story about Juniper, the gray filly on a horse farm near Ocala, Florida 1Tessa's cousin Bayo wanted to lead Juniper into a horse trailer. He pulled hard on the rope, and Juniper planted all four hooves in the dirt. 2Then Bayo set a bucket of oats inside the trailer, stepped back, and waited. After ten long minutes, Juniper walked in on her own.* Which is the BEST summary of this story? [Bayo tries to pull Juniper into a trailer, but she refuses. When he waits patiently with oats, she walks in by herself. / I think Bayo was mean to pull the rope, and this is my favorite story. / Bayo has a rope and a bucket of oats, and the trailer is near Ocala. / Juniper kicks the trailer and runs away to the river.]	**Bayo tries to pull Juniper into a trailer, but she refuses. When he waits patiently with oats, she walks in by herself.**	The best summary names the characters, the problem, the turning point, and the ending, in new words, with no opinions or tiny details. The others give an opinion, a tiny detail only, or an event that never happens.
h2_s1	Set 2: Type one word: a short retelling of only the most important ideas, in order, in your own words, is a ___. Word:	**summary** (also accepted: a summary, summaries)	A summary is a short retelling of only the most important ideas, in order, in your own words, with no opinions.
h1_s2	Set 3: *Ice and Sun The ice will snap a branch in two when winter wind blows strong, but spring will coax the buds anew with sunshine all day long.* Two steps: what does the stanza contrast, AND which theme does it show? [Harsh ice and gentle sunshine; gentleness can do what force cannot / Spring and summer; it is always sunny in spring / Branches and buds; trees grow tall / Wind and rain; storms are scary]	**Harsh ice and gentle sunshine; gentleness can do what force cannot**	The stanza contrasts harsh ice that snaps a branch with gentle sunshine that coaxes buds to open. That shows the theme that gentleness can do what force cannot.
h2_s2	Set 3: Type one word: a group of lines in a poem, like the four lines of "Ice and Sun," is called a ___. Word:	**stanza** (also accepted: a stanza, stanzas)	A group of lines in a poem is a stanza. Poems use lines and stanzas instead of paragraphs.

TITLES AND DIALOGUE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
h3_s0	Set 1: Fix this sentence: In the fable The Oak and the Reeds, the reeds bend.	**In the fable "The Oak and the Reeds," the reeds bend.**	The title of a fable goes inside quotation marks, and the comma after the title goes inside the closing quotation mark.
h3_s1	Set 2: Fix this sentence: "give her time" uncle Femi said.	**"Give her time," Uncle Femi said.**	Capitalize Give, the first word of the spoken sentence, and Uncle, part of his name. When the speaker tag comes after the words, a comma goes inside the closing quotation mark.
h3_s2	Set 3: Fix this sentence: the poem What the River Knows has four stanzas	**The poem "What the River Knows" has four stanzas.**	Capitalize the first word. The title of a poem goes inside quotation marks. End with a period.

PUTTING SKILLS TOGETHER (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Two steps: Leilani splits 2.4 kilograms of macadamia nuts equally into 6 bags. How many GRAMS go in each bag? Grams:	**400** (also accepted: 400 grams, 400 gram, 400 g)	Step 1: 2.4 ÷ 6 = 0.4 kilogram in each bag. Step 2: 1 kilogram = 1,000 grams, so 0.4 × 1,000 = 400 grams.
x2_s0	Set 1: Which explanation is correct? A jam box is 6 centimeters long, 5 centimeters wide, and 2 centimeters tall. What is its volume? [60 cubic centimeters, because each layer holds 6 × 5 = 30 cubes and there are 2 layers / 13 cubic centimeters, because 6 + 5 + 2 = 13 / 60 square centimeters, because 6 × 5 × 2 = 60 / 30 cubic centimeters, because the base is 6 × 5 = 30]	**60 cubic centimeters, because each layer holds 6 × 5 = 30 cubes and there are 2 layers**	One layer holds 6 × 5 = 30 unit cubes, and there are 2 layers: 30 × 2 = 60 cubic centimeters. Adding the edges gives a length, volume is in cubic units, and the base alone is only one layer.
x1_s1	Set 2: Two steps: 8 straight streets leave a traffic circle, evenly spaced all the way around. What is the angle between two streets that have exactly one street between them? Degrees:	**90** (also accepted: 90°, 90 °, 90 degrees, 90 degree, 90 deg)	Step 1: 8 evenly spaced streets split 360° into 8 equal angles: 360 ÷ 8 = 45°. Step 2: streets with one street between them are 2 spaces apart: 2 × 45 = 90°.
x2_s1	Set 2: Which explanation is correct for 4.5 ÷ 9? [0.5, because 45 tenths ÷ 9 = 5 tenths / 5, because 45 ÷ 9 = 5 / 0.05, because dividing always moves the decimal point two places / 2, because 9 − 4.5 is about 4.5, which is half of 9]	**0.5, because 45 tenths ÷ 9 = 5 tenths**	4.5 is 45 tenths. 45 tenths ÷ 9 = 5 tenths = 0.5. Check: 0.5 × 9 = 4.5. (5 forgets the decimal point, and 0.05 puts it one place too far.)
x1_s2	Set 3: Two steps: a cold room is 10 feet long, 8 feet wide, and 9 feet tall. Metal shelves take up 70 cubic feet. How many cubic feet are left for peach boxes? Cubic feet:	**650** (also accepted: 650 cubic feet, 650 cubic foot, 650 cubic ft, 650 cu ft, 650 ft3, 650 ft³, 650 cubic units)	Step 1: the room is 10 × 8 × 9 = 720 cubic feet. Step 2: the shelves take 70, so 720 − 70 = 650 cubic feet are left.
x2_s2	Set 3: Which explanation is correct for 1/2 ÷ 5? [1/10, because splitting a half into 5 equal parts makes tenths of the whole / 10, because 2 × 5 = 10 / 5/2, because 5 ÷ 2 = 5/2 / 1/7, because 2 + 5 = 7]	**1/10, because splitting a half into 5 equal parts makes tenths of the whole**	Splitting one half into 5 equal parts makes 10 equal parts in the whole, so each part is 1/10. Check: 1/10 × 5 = 5/10 = 1/2.

BE THE CHECKER (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x3_s0	Set 1: Find the mistake: Silas says 6 ÷ 1/3 = 2, because 6 ÷ 3 = 2. What went wrong? [He divided by 3 instead of 1/3. Each whole has 3 thirds, so 6 ÷ 1/3 = 18 / Nothing. 6 ÷ 1/3 = 2 / He should get 1/18, because 1 ÷ (6 × 3) = 1/18 / He should get 6 1/3, because you add the 1/3]	**He divided by 3 instead of 1/3. Each whole has 3 thirds, so 6 ÷ 1/3 = 18**	He found 6 ÷ 3. But 1/3 is the size of each piece: each whole has 3 thirds, so 6 wholes have 6 × 3 = 18 thirds. Check: 18 × 1/3 = 6.

ID	Question	Answer	Why (explain to student)
x4_s0	Set 1: Find the mistake: Noor turned the script line PALOMA: (whispering) Thirty-four degrees. into a story sentence: PALOMA: “Thirty-four degrees,” she whispered. What is wrong? [A story sentence drops the name label and colon: Paloma whispered, “Thirty-four degrees.” / Nothing. Stories always use NAME + colon / She should take away the quotation marks and keep the colon / She should keep (whispering) in parentheses inside the quotation marks]	**A story sentence drops the name label and colon: Paloma whispered, “Thirty-four degrees.”**	A story sentence drops the name label and colon, adds a tag, and puts the exact words in quotation marks: Paloma whispered, “Thirty-four degrees.” A script never uses quotation marks.
x3_s1	Set 2: Find the mistake: Imani measures an OBTUSE angle with a protractor and writes 55°. What went wrong? [She read the wrong scale. An obtuse angle is more than 90°, so it is 180 − 55 = 125° / Nothing. Obtuse angles can be 55° / She should add 90, so the angle is 145° / She should have used a ruler instead of a protractor]	**She read the wrong scale. An obtuse angle is more than 90°, so it is 180 − 55 = 125°**	An obtuse angle is more than 90°, so 55° cannot be right. The two protractor scales add to 180 at every mark, so she read the wrong scale: 180 − 55 = 125°.
x4_s1	Set 2: Find the mistake: Mateo writes, “Our library should stay open until eight. Everyone knows libraries are the best.” What is the BIGGEST problem? [His only support is an opinion that cannot be checked, so it is not evidence / He has no claim at all / He should never give a reason / His claim is a fact, so no one could disagree]	**His only support is an opinion that cannot be checked, so it is not evidence**	“Everyone knows libraries are the best” is an opinion that cannot be checked, so it does not prove the claim. Strong evidence is a fact, a number, an example, or an expert’s words.
x3_s2	Set 3: Find the mistake: Makena says 0.35 ÷ 5 = 0.7. What went wrong? [She left out the placeholder zero. 35 hundredths ÷ 5 = 7 hundredths, so the answer is 0.07 / Nothing. 0.35 ÷ 5 = 0.7 / The answer should be 7, because 35 ÷ 5 = 7 / She should multiply instead: 0.35 × 5 = 1.75]	**She left out the placeholder zero. 35 hundredths ÷ 5 = 7 hundredths, so the answer is 0.07**	0.35 is 35 hundredths. 35 hundredths ÷ 5 = 7 hundredths = 0.07. Checking her answer shows the mistake: 0.7 × 5 = 3.5, not 0.35.
x4_s2	Set 3: Find the mistake: a student writes that the theme of “The Wind and the Sun” is “Kindness.” What is wrong? [One word is a topic. A theme is a full sentence, such as “Being kind often works better than being harsh.” / Nothing. A theme should always be one word / The theme should be a command, like “Be nice.” / Fables do not have themes]	**One word is a topic. A theme is a full sentence, such as “Being kind often works better than being harsh.”**	One word, like Kindness, is only a topic. A theme is a full-sentence message about life, such as “Being kind often works better than being harsh.”

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Fix this sentence: leilani asked, which basket holds the papayas	**Leilani asked, "Which basket holds the papayas?"**	Capitalize the name Leilani, put quotation marks around her exact words, capitalize Which, and put the question mark inside the closing quotation mark.
s2_s0	Set 1: Fix this sentence: The orchard sells it's cider in glass jugs.	**The orchard sells its cider in glass jugs.**	Its (no apostrophe) shows ownership: the cider belongs to the orchard. It’s always means it is.

ID	Question	Answer	Why (explain to student)
s3_s0	Set 1: Fix this sentence: imani and her uncle walked down pennsylvania avenue	**Imani and her uncle walked down Pennsylvania Avenue.**	Capitalize the first word, the name Imani, and both words of the street name Pennsylvania Avenue. End with a period.
s4_s0	Set 1: Fix this sentence: The boxes of peaches in the cold room is ready.	**The boxes of peaches in the cold room are ready.** (also accepted: The boxes of peaches in the cold room were ready.)	The subject is boxes (plural), so the verb is are, not is. The words “of peaches in the cold room” do not change the subject.
s1_s1	Set 2: Fix this sentence (Paloma’s words are an exclamation): the pump is working Paloma shouted	**"The pump is working!" Paloma shouted.**	Put quotation marks around Paloma’s exact words and capitalize The. Her words are an exclamation, so the exclamation point goes inside the closing quotation mark. End the whole sentence with a period.
s2_s1	Set 2: Fix this sentence: For example the garden could grow tomatoes beans and peppers.	**For example, the garden could grow tomatoes, beans, and peppers.** (also accepted: For example, the garden could grow tomatoes, beans and peppers.)	For example is an introductory phrase, so a comma follows it. Tomatoes, beans, and peppers are three items in a series, so put a comma after each item except the last.
s3_s1	Set 2: Fix this sentence: makena lives in fort valley, georgia	**Makena lives in Fort Valley, Georgia.**	Capitalize the first word and the name Makena, both words of the town name Fort Valley, and the state Georgia. End with a period.
s4_s1	Set 2: Fix this sentence: Each of the pallets hold 32 boxes.	**Each of the pallets holds 32 boxes.** (also accepted: Each of the pallets held 32 boxes.)	The subject is Each, which means each ONE, so it is singular and needs holds, not hold. The words “of the pallets” do not change the subject.
s1_s2	Set 3: Fix this sentence: Noor said, "the barometer is dropping".	**Noor said, "The barometer is dropping."**	Capitalize The, the first word of Noor’s spoken sentence, and move the period inside the closing quotation mark.
s2_s2	Set 3: Fix this sentence: Tessa read the fable The Wind and the Sun to her cousin.	**Tessa read the fable "The Wind and the Sun" to her cousin.**	The title of a fable goes inside quotation marks.
s3_s2	Set 3: Fix this sentence: first the crew lined up the protractor on the ray	**First, the crew lined up the protractor on the ray.**	Capitalize the first word, and put a comma after the introductory word First. End with a period.
s4_s2	Set 3: Fix this sentence: The mangoes and the pineapple was on sale.	**The mangoes and the pineapple were on sale.** (also accepted: The mangoes and the pineapple are on sale.)	“The mangoes and the pineapple” is a compound subject joined by and, so it needs the plural verb were, not was.

WRITER'S NOTEBOOK: MAKE YOUR CASE (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k1_s0	Set 1: Which sentence uses a comma after an introductory phrase AND gives evidence (a fact), not an opinion? [For example, the Kaka'ako Farmers Market is open every Saturday morning. / For example the Kaka'ako Farmers Market is open every Saturday morning. / For example, farmers markets are the most fun place on Earth. / The Kaka'ako Farmers Market is open every Saturday morning.]	**For example, the Kaka'ako Farmers Market is open every Saturday morning.**	It has a comma after the introductory phrase For example, and it gives a fact that can be checked. One choice is missing the comma, one is an opinion, and one has no introductory phrase.
k1_s1	Set 2: Which sentence correctly punctuates dialogue AND includes a fact, not an opinion? ["A pineapple takes about two years to grow," Leilani said. / "A pineapple takes about two years to grow" Leilani said. / "Pineapples are the best fruit in the world," Leilani said. / A pineapple takes about two years to grow, "Leilani said."]	**"A pineapple takes about two years to grow," Leilani said.**	The comma sits inside the closing quotation mark before the tag, and the words give a checkable fact from the lesson: a pineapple takes about two years to grow. The others are missing the comma, give an opinion, or put the marks around the tag.
k1_s2	Set 3: Which sentence states a CLAIM and uses commas in a series correctly? [Our town should plant a garden with tomatoes, beans, and squash. / Our town should plant a garden with tomatoes beans, and squash. / The garden has tomatoes, beans, and squash. / Our town should plant a garden, with tomatoes, beans and, squash.]	**Our town should plant a garden with tomatoes, beans, and squash.**	It states a claim (an opinion people could disagree with), and it puts a comma after each item in the series except the last. One choice is missing a comma, one is a fact instead of a claim, and one puts commas in the wrong places.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match unit fraction	**a fraction with a numerator of 1**	2 points if matched on the first try.
match_s0_2	Set 1: Match quotient	**the answer to a division problem**	2 points if matched on the first try.
match_s0_3	Set 1: Match perpendicular lines	**lines that meet at right angles**	2 points if matched on the first try.
match_s0_4	Set 1: Match stage directions	**notes that tell actors how to move or speak**	2 points if matched on the first try.
match_s0_5	Set 1: Match counterclaim	**the other side's opinion in an argument**	2 points if matched on the first try.
match_s1_1	Set 2: Match placeholder zero	**a zero that holds an empty place**	2 points if matched on the first try.
match_s1_2	Set 2: Match vertex	**the point where an angle's two rays meet**	2 points if matched on the first try.
match_s1_3	Set 2: Match volume	**the amount of space inside a solid figure**	2 points if matched on the first try.
match_s1_4	Set 2: Match prefix	**a word part added to the front of a word**	2 points if matched on the first try.

ID	Question	Answer	Why (explain to student)
match_s1_5	Set 2: Match theme	**a message about life, written as a full sentence**	2 points if matched on the first try.
match_s2_1	Set 3: Match divisor	**the number you divide by**	2 points if matched on the first try.
match_s2_2	Set 3: Match obtuse angle	**more than 90° but less than 180°**	2 points if matched on the first try.
match_s2_3	Set 3: Match composite figure	**a solid made of two or more prisms**	2 points if matched on the first try.
match_s2_4	Set 3: Match evidence	**facts or numbers that prove a reason**	2 points if matched on the first try.
match_s2_5	Set 3: Match summary	**a short retelling of only the most important ideas**	2 points if matched on the first try.

UNIT 4 BOSS CHALLENGE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
z1_s0	Set 1: *The Situation At the packing house in Fort Valley, Makena fills a wooden bin that is 4 feet long, 3 feet wide, and 2 feet tall with peaches. Then the peaches are packed into small boxes that each hold 1/4 cubic foot. Later, Makena writes a script for her class play about the day.* Two steps: how many 1/4-cubic-foot boxes can be filled with the peaches from the full bin? Boxes:	**96** (also accepted: 96 boxes, 96 box)	Step 1: the bin holds 4 × 3 × 2 = 24 cubic feet. Step 2: 24 ÷ 1/4 = 24 × 4 = 96 boxes. Check: 96 × 1/4 = 24.
z2_s0	Set 1: Makena's script says: MAKENA: (proudly) We packed ninety-six boxes! Which story sentence turns it into prose correctly AND keeps the stage direction's meaning? [Makena said proudly, "We packed ninety-six boxes!" / MAKENA: "We packed ninety-six boxes!" she said proudly. / Makena said proudly, "we packed ninety-six boxes"! / Makena said proudly, We packed ninety-six boxes!]	**Makena said proudly, "We packed ninety-six boxes!"**	A story sentence drops the name label, keeps the meaning of the stage direction (proudly) in the tag, puts a comma after the tag, and keeps the exclamation point inside the closing quotation mark. The others keep the script label, put the end mark outside, or leave out the quotation marks.
z1_s1	Set 2: *The Situation At the Honolulu farmers market, Mr. Haddad has 14.4 liters of pineapple juice. He pours it equally into 6 pitchers, then pours each pitcher equally into 8 cups. Leilani is writing a letter to the market manager.* Three steps: how many MILLILITERS of juice are in each cup? (1 liter = 1,000 milliliters) Milliliters:	**300** (also accepted: 300 milliliters, 300 milliliter, 300 mL, 300 ml)	Step 1: 14.4 ÷ 6 = 2.4 liters in each pitcher. Step 2: 2.4 ÷ 8 = 0.3 liter in each cup. Step 3: 1 liter = 1,000 milliliters, so 0.3 × 1,000 = 300 milliliters.

ID	Question	Answer	Why (explain to student)
z2_s1	Set 2: Leilani's draft: First the market should stay open until two o'clock. Last month, 40 customers arrived after noon. Which revision fixes the comma AND keeps the evidence? [First, the market should stay open until two o'clock. Last month, 40 customers arrived after noon. / First, the market should stay open until two o'clock. Everyone loves shopping late. / First the market should stay open until two o'clock. Last month, 40 customers arrived after noon. / First, the market, should stay open until two o'clock. Last month, 40 customers arrived after noon.]	**First, the market should stay open until two o'clock. Last month, 40 customers arrived after noon.**	First is an introductory word, so it needs a comma, and the second sentence keeps the checkable evidence (40 customers). One choice replaces the evidence with an opinion, one is still missing the comma, and one adds a comma in the wrong place.
z1_s2	Set 3: *The Situation In Washington, D.C., a diagonal avenue meets a straight street. On the north side of the street, the avenue makes a 48° angle with the street. Ms. Ruiz splits the other angle on the north side into 4 equal parts with new bike lane lines. Meanwhile, near Ocala, Tessa is writing about the story "Juniper in the Tall Grass."* Two steps: what does each of the 4 equal parts measure? Degrees:	**33** (also accepted: 33°, 33 °, 33 degrees, 33 degree, 33 deg)	Step 1: the two angles on the north side make a straight angle: 180 − 48 = 132°. Step 2: split into 4 equal parts: 132 ÷ 4 = 33°. Check: 4 × 33 = 132, and 132 + 48 = 180.
z2_s2	Set 3: Tessa wants to prove the theme Patience can do what force cannot with a quote from the story. Which sentence quotes correctly AND supports that theme? [The author writes, "Slowly, without standing up, Tessa raised the halter, and the filly let it slide over her ears." / The author writes, Slowly, without standing up, Tessa raised the halter, and the filly let it slide over her ears. / The author writes, "Her uncle Femi was leaning on a post with a mug of coffee." / The author writes, "Slowly, without standing up, Tessa raised the halter, and the filly let it slide over her ears".]	**The author writes, "Slowly, without standing up, Tessa raised the halter, and the filly let it slide over her ears."**	This quote copies the story's exact words inside quotation marks, with a comma after the tag and the period inside, and it shows patience working. One choice has no quotation marks, one quotes a detail that does not support the theme, and one puts the period outside.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade4-5/prepost/KCE_Checkpoint_PrePost_G4-5W20.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	1/3 ÷ 2 and 5 ÷ 1/4; check one	(a) 1/6 (b) 20. Check: 1/6 × 2 = 2/6 = 1/3, or 20 × 1/4 = 5.
2	6.6 ÷ 3 and 47.5 ÷ 100	(a) 2.2 (b) 0.475
3	Two angles on a straight line; one is 140°	180° − 140° = 40°. The 140° angle is obtuse.
4	Volume of a 5 × 3 × 4 inch box	60 cubic inches
5	A word with the root port (carry)	Answers vary: portable (able to be carried), transport (carry across), import, export, porter.

#	Prompt	Model answer / grading note
6	Sentence with correct commas	However, the lot needs seeds, soil, and tools.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	7.2 L into 4 pitchers, then each into 6 cups	7.2 ÷ 4 = 1.8 liters per pitcher; 1.8 ÷ 6 = 0.3 liter per cup.
2	Makena says a 6 × 2 × 3 box is 11 cubic feet	She added the edges (6 + 2 + 3). Volume = 6 × 2 × 3 = 36 cubic feet.
3	Be the editor	a) Paloma said, "We have to save the berries." b) First, the garden would need hoses, rakes, and seeds. c) The boxes on the truck are cold.
4	Writing: patience or force?	Grading note: a clear claim; a reason starting “First,”; evidence from a Unit 4 text (for example, the Sun in the fable or Tessa with Juniper); one correctly punctuated line of dialogue; 4–5 complete sentences.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	3 pounds into 1/4-pound bags; 3 bags per gift box	3 ÷ 1/4 = 12 bags; 12 ÷ 3 = 4 gift boxes.